AXI4-Stream to PCIe 6.0-Style Link: A Synthesizable RTL Bridge with FLIT Framing, FEC/CRC Protection and Replay

Project report – SEMICON problem statement: RTL for AXI4 (Master & Slave) to PCIe 6 Link

Abstract

SmartNICs and machine-learning accelerators produce continuous AXI4-Stream traffic that must reach host memory over PCIe. This report describes a synthesizable RTL bridge that accepts traffic from N AXI4-Stream masters, arbitrates between them with packet-atomic QoS scheduling, packs the traffic into 256-byte PCIe 6.0-style FLITs protected by an 8-byte CRC and a 6-byte FEC, tracks them with sequence numbers, ACK/NAK control FLITs and a go-back-N replay buffer, and stripes each FLIT across four lanes with scrambling, Gray coding and PAM4 precoding. A matching receive path decodes the lanes, corrects single-symbol errors, checks the CRC and delivers the original records, including per-byte keep/strobe and routing metadata, on an AXI-Stream-style output. A testbench with two back-to-back instances, a scoreboard and error injection exercises network-packet and ML-inference workloads. The complete design was synthesized with Cadence Genus against the `tsl18fs120_scl_ss` standard-cell library with a 10 ns (100 MHz) clock: the netlist has 203,069 cells (50,042 flip-flops) and a cell area of 6,505,672.955 area units, all 40 reported setup paths meet timing (worst slack 0 ps, TNS 0), and the estimated power is 542.3 mW. These are pre-layout synthesis results; place-and-route and GDSII are future work. The report states clearly which parts are modelled on, rather than identical to, the licensed PCIe 6.0 specification.

Keywords: AXI4-Stream, PCIe 6.0, FLIT, forward error correction, CRC, replay buffer, PAM4, QoS arbitration, RTL, logic synthesis.

1 Introduction and Problem Statement

1.1 Use cases

An AXI4 to PCIe link serves several workloads. The problem statement names two:

- **Network packet processing.** SmartNICs and networking hardware process continuous AXI-Stream traffic such as Ethernet packets. The conversion hardware bridges these streams into PCIe so they reach host server memory directly.
- **Machine-learning inference accelerators.** AI acceleration cards (for example FPGA-based cards) take a stream of processed neural-network weights or pixel data and send it directly into a host memory buffer.

The two workloads stress the bridge differently. Network traffic is made of many packets of very different lengths whose last beat is usually only partly filled. ML traffic is made of long, back-to-back bursts that need sustained throughput, together with short read requests.

1.2 Objective

Develop a synthesizable RTL design that follows the given block diagram: AXI4-Stream (512-bit) inputs, an interconnect layer (central crossbar and dynamic arbiters), a link layer (packetizer and

error correction with FEC and CRC, bundling data into fixed 256-byte FLITs), and a protocol-to-PHY layer (multi-lane striping over lanes 0 to 3, Gray coding and PAM4 mapping, 2 bits to 1 symbol) that drives a 4-lane symbol bus towards an analog PHY and SerDes.

1.3 Deliverables and tools

- Complete synthesized RTL and GDSII using OpenROAD/OpenLane (the data rate may be reduced depending on the technology node).
- Exhaustive functional test cases relevant to the two use cases.
- Bonus: FPGA-tested AXI4 to PCIe IP.
- Tools: Python, Verilog/SystemVerilog/SVA, Icarus Verilog, Xilinx Vivado (host-controller IPs from Xilinx may not be used), OpenROAD or OpenLane with open-source PDKs.

Scope of this report. The RTL, its testbench and the logic synthesis of the complete design (Cadence Genus, Section 8) are complete and are described here. Place-and-route, GDSII generation (OpenLane/OpenROAD) and the FPGA bonus have not been run in this work and are listed as future work (Section 9).

1.4 Contributions

1. A packet-atomic N:1 QoS interconnect that never interleaves beats of different packets.
2. A lossless metadata scheme: a 24-byte record header carries every AXI-Stream sideband signal, and a 6-byte DLP carries link-level framing and sequencing.
3. A reliable FLIT link: CRC, 3-way interleaved FEC with single-symbol correction, sequence numbers, ACK/NAK and go-back-N replay.
4. A PHY mapping with byte striping across four lanes, scrambling, Gray coding and PAM4 precoding, with the inverse on receive.
5. A self-checking two-instance testbench with error injection.
6. A complete logic-synthesis run of the full-duplex design with area, power and timing results and a design-rule check, which identifies the PHY scrambler/precoder as the timing-critical block.

2 Background

2.1 AXI4-Stream signals

AXI4-Stream is a point-to-point streaming protocol. A transfer occurs in a clock cycle in which both TVALID and TREADY are high. Packets are delimited by TLAST. The table lists the signals used here. The last three rows are project sideband extensions, not part of the AXI4-Stream standard.

Signal	Width	Meaning
tdata	512	Payload bytes of the beat
tkeep	64	One bit per byte of tdata; 0 marks a null byte that carries no information
tstrb	64	One bit per byte; distinguishes data bytes from position bytes
tid	12	Identifies the source stream (here: master index and packet number in the testbench)
tdest	8	Routing information for the destination
tuser	8	User-defined sideband
tlast	1	Marks the final beat of a packet
tdrw (extension)	1	1 = write/data beat, 0 = read request (header only)
address (extension)	31	Target host-memory address of the beat
tqos (extension)	7	Priority of the source; used only for arbitration (Section 5.2)

2.2 PCIe 6.0 concepts modelled

- **FLIT mode.** PCIe 6.0 moves data in fixed-size 256-byte FLITs. We use 236 bytes of payload, 6 bytes of DLP, 8 bytes of CRC and 6 bytes of FEC.
- **PAM4 signalling.** Each symbol carries 2 bits. Gray coding and precoding limit the error burst length caused by a single symbol error.
- **FEC plus CRC.** A lightweight, 3-way interleaved FEC corrects small errors without a retry; a CRC detects what FEC cannot correct, and the link layer recovers by replay.
- **Replay.** Flits carry sequence numbers; the receiver returns ACK or NAK and the transmitter retransmits from a replay buffer.

The PCIe 6.0 specification is licensed, so some numerical details (CRC first-consecutive-root, FEC symbol structure, scrambler polynomial) are modelled assumptions that follow public descriptions. They are collected in Section 9.

3 Our Approach

3.1 Starting point

We kept the three-layer structure of the given block diagram (interconnect, link layer, protocol-to-PHY) and implemented each layer as a separate set of modules with a narrow interface between layers: a record stream between the interconnect and the link layer, a 256-bit beat stream between the link layer and the PHY layer, and four 64-bit lane buses to the analog side.

3.2 Methodology

1. Implement a baseline of each layer and a smoke-test testbench.
2. Review the baseline against the use cases and against PCIe concepts, and list gaps.
3. Close each gap in RTL and add a directed test or error-injection test for it.
4. Extend the design from a transmit-only path to a full-duplex link so that read data, ACK/NAK and error recovery can be demonstrated.

5. Synthesize the complete design, check the netlist, and review area, power and timing to find the blocks that limit the clock rate.

3.3 Gaps found in the first version and how they were closed

Gap in the first version	Consequence	Change in the current design
Arbiter switched masters every beat and ignored tlast	Beats of two masters could interleave inside one FLIT; a tlast from either master flushed a mixed FLIT	Arbitration only at packet boundaries, grant locked until tlast, round-robin among equal priorities
tkeep, tstrb, tid, tdest, tuser were not carried; only OR-reduced keep/strb bits and the first id survived	A partial last beat (any real Ethernet frame) could not be reconstructed; a zero-padded flush had no length	24-byte header per record carries all fields; the DLP carries record count and used payload bytes
Reads carried 512 dummy data bits; no receive or completion path	Wasted bandwidth; nothing returned to the requester	A read is a header-only record (24 bytes instead of 88); full receive path with record output
No sequence number, ACK/NAK or replay	No recovery from errors although the DLP had reserved bits	10-bit sequence number in the DLP, ACK/NAK control FLITs, replay buffer
Lane mapping used contiguous 64-bit blocks; no scrambling or precoding	Not byte striping as in PCIe; lane order looked reversed	Byte j goes to lane (j mod 4); per-lane scrambler, Gray coding and mod-4 precoder, with inverse on receive
CRC and FEC existed only as generators	Error correction could not be demonstrated, only that parity bits are produced	Receive decoder: single-symbol correction per FEC group, CRC check, error-injection tests
Address was read live from the master pins rather than stored with the beat	The address could belong to a later beat	The address travels with the beat through the slave interface (packed next to tuser)

4 System Architecture

4.1 Block diagram

Figure 1 shows the architecture. Traffic from N AXI4-Stream masters enters per-master slave interfaces, is arbitrated by the interconnect, is packed and protected by the link layer, and leaves through the protocol-to-PHY layer as four lanes of PAM4 symbols. The receive path mirrors the transmit path in the opposite direction and returns records on an AXI-Stream-style output; ACK and NAK control FLITs travel on the reverse link.

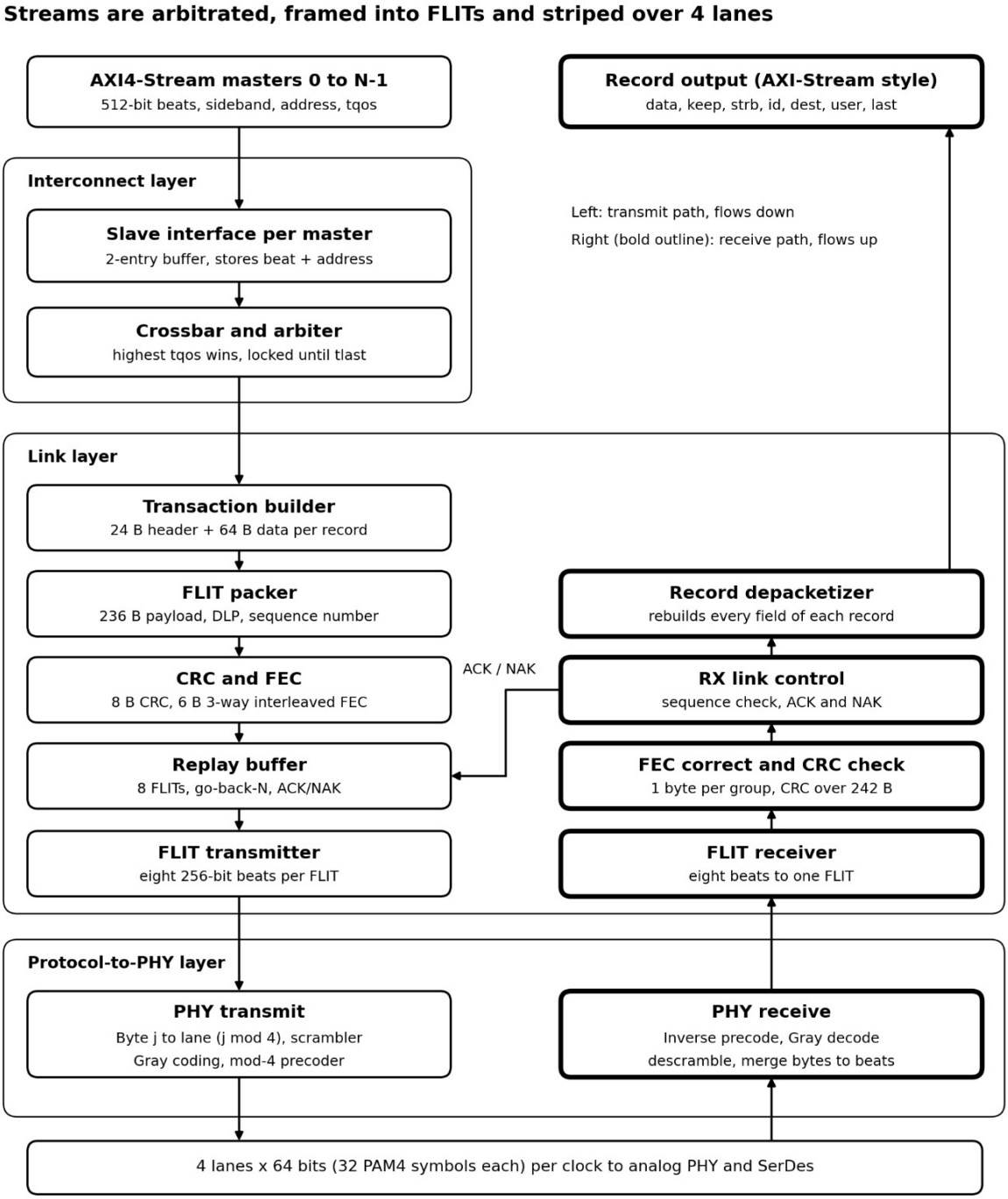


Figure 1: Architecture block diagram: three layers, transmit and receive paths

Transmit flows down the left column from the masters to the lanes; receive flows up the right column to the record output, and RX link control returns ACK and NAK to the replay buffer.

4.2 Module hierarchy

Module	Source file	Role
pcie6_top	pcie6_top.v	Integrates N slave interfaces, the interconnect, the link layer and the PHY layer (Tx and Rx)
axi4s_slave_if	axi4s_slave_if.v	Per-master two-entry elastic buffer (skid buffer)
axi_interconnect	axi_interconnect.v	N:1 crossbar with QoS arbitration and packet locking
pcie_link_layer	pcie6_link_layer.v	Tx: transaction_builder, flit_packer, replay_tx. Rx glue: rx path wiring and statistics
flit_protect, pcie6_crc_pipeline, pcie6_fec_flit, flit_transmitter	pcie6_flit_protect.v	CRC and FEC generation; FLIT to eight 256-bit beats
flit_receiver, flit_check, rx_link_ctl, rec_depack	pcie6_rx.v	Beats to FLIT, FEC correction and CRC check, sequence and ACK/NAK logic, record unpacking
pcie6_proto_to_phy, pcie6_phy_to_proto	pcie6_phy.v	Byte striping, scrambling, Gray coding, precoding and their inverses
tb_pcie6_link	tb_pcie6_link.v	Two-instance self-checking testbench

4.3 Default parameters

Parameter	Default	Meaning
N	2	Number of AXI4-Stream masters
DATA_W / KEEP_W	512 / 64	Beat width in bits and in bytes
ID_W / DEST_W / USER_W	12 / 8 / 8	Widths of tid, tdest, tuser
QW	7	Width of tqos
REPLAY_LOG2	3	Replay buffer and window size: $2^3 = 8$ FLITs
REPLAY_TMO	2048	Clocks without ACK progress before replay
FLUSH_TMO	16	Idle clocks before a partly filled FLIT is sent
NAK_TMO / ACK_EVERY / ACK_TMO	256 / 2 / 16	NAK re-arm time, ACK coalescing count, ACK delay limit
PRECODE / SCRAMBLE	1 / 1	Enable PAM4 precoding and scrambling

4.4 Life of a write packet

1. A master presents beats; the slave interface stores each beat with its address and metadata.

2. At a packet boundary the interconnect picks the master with the highest tqos (ties go round-robin) and stays locked to it until the beat with tlast.
3. The transaction builder turns each beat into a record: 24-byte header plus 64 bytes of data.
4. The FLIT packer places records into a 236-byte payload and emits a FLIT with a DLP holding type, sequence number, record count and used bytes.
5. CRC (8 bytes) and FEC (6 bytes) are added and the FLIT is copied into the replay buffer.
6. The transmitter sends the FLIT as eight 256-bit beats; the PHY layer stripes, scrambles, Gray-codes and precodes them onto four lanes.
7. The receiver reverses the PHY steps, corrects and checks the FLIT, accepts it if its sequence number is the expected one, returns an ACK and unpacks the records onto the AXI-Stream-style output.

5 Detailed Design

5.1 AXI4-Stream slave interface

Each master has its own slave interface. It is a two-entry elastic buffer: a main register stage plus a skid register. TREADY is derived only from registered state, so there is no combinational path from the master into the fabric, and the interface can accept one beat per clock without bubbles. The full beat (data, keep, strobe, tid, tdest, tuser, tlast, tdrw and the 31-bit address) is stored together, so the address and metadata always belong to the beat they describe. In the top level the address is concatenated with tuser into one 39-bit sideband field, which lets the same interface and interconnect carry it without extra ports.

5.2 Interconnect: tqos, arbitration and packet locking

What tqos is. tqos is a 7-bit unsigned priority supplied with each master (0 to 127). A larger value means a more urgent source. It is a project sideband, in the spirit of the AxQOS signals of full AXI4, and is used only inside the interconnect. It is not transmitted over the link.

Why packets must be atomic. The link layer packs records back to back and treats the tlast of a record as the end of a packet. If beats of two masters interleave, one master's tlast ends a FLIT that also holds another master's half-finished packet, and the receiver cannot tell the packets apart. The interconnect therefore arbitrates only at packet boundaries.

Arbitration rule.

- While no master is locked, the arbiter looks at all masters with a valid beat. It scans them in rotated order starting at a round-robin pointer and chooses the one with the highest tqos; the first one in scan order wins a tie.
- When a granted beat is accepted and is not the last beat, the interconnect locks to that master. The lock is released when a beat with tlast is accepted.
- When the last beat is accepted, the round-robin pointer moves to the master after the winner, so equal-priority masters alternate.

The output is a pure multiplexer of the slave-interface registers. A one-cycle grant strobe tells the winning slave interface that its beat was consumed. The output carries a read/write indication (ddrw, with decoded dwrite and dread) taken from the selected master's tdrw bit.

5.3 Transaction builder and the TLP-style record header

The transaction builder adds one register stage and forms a record for every beat. A record is a 192-bit (24-byte) header, followed by 64 bytes of data only when the beat is a write. Read requests are therefore only 24 bytes. Records are never split across FLITs. The header layout is given in Section 6.

5.4 FLIT packer and the DLP

The packer places records into the 236-byte payload accumulator. A write record needs 88 bytes and a read record needs 24 bytes. A record is accepted if it still fits. A FLIT is emitted when the next record does not fit, when not even a read record could fit, or when FLUSH_TMO idle clocks have passed with data waiting. Each emitted FLIT gets the next sequence number, and the packer stalls when the number of unacknowledged FLITs reaches the replay window. The DLP (Section 6) records the FLIT type, sequence number, number of records and number of used payload bytes. The last two fields remove the ambiguity of a zero-padded partial flush.

5.5 CRC and FEC

CRC. An 8-symbol Reed-Solomon-style check over GF(2^8) with primitive polynomial 0x11D. The generator is the product of $(x+\alpha^{FCR+i})$ for $i = 0$ to 7, and the CRC is $M(x)\,x^8 \bmod g(x)$, where M is the 242-byte message made of the 236-byte payload and the 6-byte DLP. The hardware consumes 32 bytes per clock using look-ahead ($R_{new} = AR \oplus B \cdot \text{chunk}$) in a three-stage pipeline; the 242-byte message is front-padded with 14 zero bytes to make eight chunks. The first-consecutive-root FCR is a placeholder (set to 0) that must be set from the specification.

FEC. The 250 bytes covered (message plus CRC) are split into three interleaved groups by byte index modulo 3. For group g and position k within the group the transmitter computes two check symbols: P_g, the sum of the bytes, and C_g, the sum of each byte multiplied by α^k. The six check bytes form the FEC field. The receiver recomputes them, derives a syndrome, and can locate and correct one wrong byte per group. Two wrong bytes in the same group exceed the correction capability and are left to the CRC and replay.

5.6 Reliability: sequence numbers, ACK/NAK and replay

Data FLITs carry a 10-bit sequence number in the DLP. The transmitter stores each new data FLIT in an 8-entry replay buffer.

- ACK is cumulative and frees buffer slots. The receiver coalesces ACKs (every 2 FLITs or after 16 clocks).
- NAK(n) acknowledges everything before n and makes the transmitter replay from n (go-back-N). A NAK is not overwritten by a later ACK while it is pending.
- Replay timeout. If the oldest FLIT is not acknowledged within REPLAY_TMO clocks, the transmitter replays from the oldest unacknowledged FLIT.
- Control FLITs (DLP type ACK or NAK) have an empty payload, are protected by the same CRC and FEC, and are forwarded without being stored for replay.

Received condition	Action
Data FLIT, sequence = expected	Accept, ACK (coalesced)
Data FLIT, sequence ahead of expected	Drop, NAK(expected), re-armed after NAK_TMO
Data FLIT, sequence behind expected	Drop, immediate ACK (duplicate after a replay)
CRC failure or uncorrectable FEC	Drop, NAK(expected)
Control FLIT	Forward to the replay logic

5.7 Receive path

1. flit_receiver assembles eight 256-bit beats into one FLIT and discards it with an error pulse if rx_last does not coincide with the eighth beat.
2. flit_check corrects up to one wrong byte per FEC group, then verifies the CRC over bytes 0 to 241. A FLIT is good if the CRC matches and no group was uncorrectable. If all slots are busy the FLIT is dropped and recovered later by NAK or replay.
3. rx_link_ctl applies the sequence rules below.
4. rec_depack unpacks the records of an accepted payload one per clock onto the AXI-Stream-style output, rebuilding tdata, tkeep, tstrb, tid, tdest, tuser, tlast, the read/write bit and the address.

5.8 Protocol-to-PHY layer

The PHY layer carries 256 bits per clock on four lanes of 64 bits, that is 32 PAM4 symbols per lane per clock.

- **Byte striping.** Byte j of a beat goes to lane (j mod 4), slot $j/4$. The first byte of a lane word sits in bits [63:56], and symbol 0 is sent first.
- **Scrambling.** A 23-bit LFSR per lane (polynomial $x^{23} + x^{21} + x^{16} + x^8 + x^5 + x^2 + 1$) whitens the data. It is reseeded, and the precoder cleared, at the first beat of every FLIT.
- **Gray coding.** Each 2-bit symbol $g = \{b_1,\ b_1 \oplus b_0\}$, so adjacent PAM4 levels differ by one bit.
- **Precoding.** $p[t] = (g[t] - p[t-1]) \bmod 4$, which limits an error burst after a symbol error.

The receiver applies the inverse of each step.

6 Metadata: What We Carry and Why

6.1 Where each input signal goes

Input	Carried in	Notes
tdata	Data part of a write record	64 bytes; absent for reads
tkeep	Record header, 64 bits	Per-byte valid mask is preserved
tstrb	Record header, 64 bits	Per-byte strobe is preserved
tid, tdest, tuser	Record header (12, 8, 8 bits)	Preserved for every record
tlast	Record header, 1 bit	Packet boundary survives the link
tdrw	Record header, bit 191	Distinguishes write from read request
address	Record header, 31 bits	Host-memory target
tqos	Not carried	Used only for arbitration

6.2 Why we added a TLP-style header

In PCIe, every transaction layer packet (TLP) starts with a header that says what the packet is, where it goes and which bytes it covers, and optionally carries data. Our 24-byte record header plays this role. Without it, the receiver would see only 512-bit data blocks and could not rebuild the AXI-Stream beat: it would not know whether the beat was a read or a write, which bytes of a partial last beat were valid, which packet it belonged to, or where the packet ended. Putting the metadata in the header per record fixes this for every beat, not just for the first beat of a FLIT. It also lets a read request be sent as a header only, which saves 64 bytes per read.

A write record is $24 + 64 = 88$ bytes; a read record is 24 bytes.

Bits	Field	Meaning
[191]	rw	1 = write (64 bytes of data follow), 0 = read request
[190]	last	tlast of the source beat
[189:159]	address	31-bit target address
[158:147]	id	tid, 12 bits
[146:139]	dest	tdest, 8 bits
[138:131]	user	tuser, 8 bits
[130:67]	keep	tkeep, 64 bits
[66:3]	strb	tstrb, 64 bits
[2:0]	reserved	Zero

6.3 Why we added a DLP

The data link layer packet (DLP) holds information about the link rather than about a transaction. Our records have variable sizes (88 or 24 bytes) and FLITs may be sent partly filled after a timeout, so the receiver needs to know how many records and bytes are valid. Reliability needs a FLIT type and a sequence number, and ACK/NAK must travel on the same protected link. Keeping this in a

separate 6-byte field that is covered by the CRC and FEC keeps the transaction metadata and the link control apart.

Bits	Field	Meaning
[47:46]	type	00 data FLIT, 01 ACK, 10 NAK
[45:36]	seq	Data FLIT: its sequence number. ACK/NAK: next sequence number expected
[35:32]	nrec	Number of records in the payload (data FLITs)
[31:24]	nbytes	Number of used payload bytes (data FLITs)
[23:0]	reserved	Zero, available for future use

6.4 FLIT layout (256 bytes)

The transmitter sends a FLIT as eight 256-bit beats, MSB first, so a FLIT costs eight clocks.

Bits	Size	Field
[2047:160]	236 B	Payload (records, zero-padded after nbytes)
[159:112]	6 B	DLP
[111:48]	8 B	CRC over payload and DLP (242 bytes)
[47:0]	6 B	FEC over payload, DLP and CRC (250 bytes)

7 Verification

7.1 Environment

The testbench instantiates two copies of pcie6_top, A and B, connected back to back so that both directions of the link are exercised. Each side has two AXI-Stream masters. Packets carry random data; the source is encoded in tid (bit 0 is the master index, bits 11:4 the packet number) so the receiver can attribute every beat. A scoreboard stores every beat sent and compares every beat that appears on the record output of the other side: read/write bit, data (writes), tkeep, tstrb, tuser, tid, tdest, tlast and address. It also checks that packets do not interleave at the output and that no extra or missing beats occur. The design exports statistics counters (transmitted FLITs, replays, received FLITs, FEC corrections, CRC failures, drops) that the tests use. Random backpressure can be applied on the record outputs.

7.2 Error injection

Symbols on the A-to-B lanes can be flipped with per-lane XOR masks applied to a chosen valid beat, and every beat on the B-to-A lanes can be corrupted to destroy the ACK path. This produces correctable errors, uncorrectable errors and lost ACKs.

7.3 Test plan

Test	Scenario	What is checked	Workload
T1	Reset and idle	No activity while idle	Basic
T2	Single 64-byte write	Data, metadata and address arrive intact	Basic
T3	Read request	Header-only record, rw = 0	ML read path
T4	Two masters, 6-beat packets at once	Packet locking, QoS order, no interleaving	Both
T5	Partial last beat (13 bytes and 1 byte)	tkeep/tstrb reconstructed exactly	Network
T6	4 sources × 12 packets of 1 to 24 beats with some reads (up to 1536 bytes, above the 1518-byte Ethernet frame)	Mixed lengths and directions, partial last beats	Network packet processing
T7	64-beat weight tensor plus 10 read requests, with a 32-beat completion stream in the reverse direction	Sustained bursts, reads and returned data	ML inference
T8	One symbol error	FEC corrects it: correction counter rises, no CRC failure, no replay	Error recovery
T9	Two bytes wrong in one FEC group	CRC failure, NAK, replay, data still correct	Error recovery
T10	ACK path corrupted for about 1800 clocks	Window stall, replay timeout, recovery	Error recovery
T11	Random backpressure on the outputs	No data loss or duplication	Both
T12	Reset in operation	Clean restart, counters cleared	Basic

7.4 Results

[To be completed from the ModelSim transcript: paste the per-test PASS/FAIL lines, the number of beats checked, the error count and the final summary line. Also add the statistics counter values printed before the reset test, and one waveform screenshot of a T2 write and a T9 replay.]

7.5 Further verification planned

A Python golden model of the FLIT framing, CRC and FEC for cross-checking the RTL, SVA assertions on the interfaces (valid held until ready, no packet interleaving, sequence window never exceeded), constrained-random traffic with coverage, and a gate-level simulation of the synthesized netlist (`pcie6_mapped_opt.v`) or an FPGA run.

8 Physical Design: Logic Synthesis

This section reports the logic synthesis of the complete full-duplex design (top module `pcie6_top`, transmit and receive paths). All figures below are taken directly from the reports generated by the synthesis tool (area, power, QoR, timing and design-check reports) and from the mapped netlist. Place-and-route, clock-tree synthesis and GDSII generation have not been run, which limits how these numbers should be read (Section 8.7).

8.1 Implementation tools and technology

Item	Value
Synthesis tool	Cadence Genus Synthesis Solution 20.11-s111_1
Technology library	`tsl18fs120_scl_ss`; the name indicates a 180 nm-class standard-cell library at a slow-slow (ss) corner (node inferred from the library name)
Operating conditions	`_nominal_` (balanced_tree)
Wireload model	Enclosed mode; default wireload of the library (no layout information)
Area mode	Timing library
Top module	`pcie6_top`
Clock	One clock domain, named `HCLK` in the constraints, period 10,000 ps (100 MHz), ideal
Output netlist	`pcie6_mapped_opt.v` (gate-level Verilog, six modules, about 502 thousand lines)
Synthesis run (date)	23 October 2026
Runtime	5,272 s CPU, 6,055 s elapsed (about 1 h 41 min)
Peak memory	4,313.65 MB (Genus)

The tool kept five sub-blocks as separate hierarchies (the packer, the Tx and Rx CRC pipelines, and the Tx and Rx PHY blocks) and flattened the rest of the design into `pcie6_top`. The mapped netlist therefore contains six modules.

What the clock rate means for the data rate. The PHY layer moves 256 bits per clock on four 64-bit lanes (32 PAM4 symbols per lane per clock). At 100 MHz this is a raw line rate of 256×100 MHz $= 25.6$ Gb/s, that is 6.4 Gb/s (3.2 GBd) per lane, and at most $236/256 \times 25.6 \approx 23.6$ Gb/s of FLIT payload before record-header overhead. A full-speed PCIe 6.0 lane runs at 64 GT/s, so this implementation runs at about one tenth of the per-lane rate. The problem statement allows the data rate to be reduced depending on the technology node.

8.2 Summary of results

Metric	Result
Clock period / frequency	10,000 ps / 100 MHz
Setup timing	Worst slack 0 ps, total negative slack 0, 0 violating paths
Total cell area	6,505,672.955 (library area units; about 6.51 mm^2 if the unit is μm^2)
Leaf instances	203,069 (50,042 sequential, 153,027 combinational)
Hierarchical instances	5
Total power (activity-based)	542.259 mW (leakage 76.97 μW, internal 457.011 mW, switching 85.171 mW)
Design check	No unresolved references, empty modules, assigns, multi-driven nets or undriven ports; 3 undriven hierarchical pins (unused)
Average fanout / max fanout	3.0 / 50,042 (the clock net)

8.3 Area

The area report lists the sub-blocks that the tool kept as hierarchy. The remaining logic was flattened into the top module, so the report gives it only as a remainder.

Instance	Module	Cells	Cell area	Share
`u_link_layer_u_packer`	`flit_packer_WIN8_FLUSH_TMO16`	13,458	470,968.722	7.24%
`u_link_layer_u_protect_u_crc`	`pcie6_crc_pipeline (Tx CRC)`	5,759	150,179.091	2.31%
`u_link_layer_u_check_u_crc`	`pcie6_crc_pipeline (Rx CRC)`	5,756	150,157.151	2.31%
`u_proto_to_phy`	`pcie6_proto_to_phy`	3,903	99,921.841	1.54%
`u_phy_to_proto`	`pcie6_phy_to_proto`	2,881	79,160.130	1.22%
Sub-blocks listed above		31,757	950,386.935	14.61%
Rest of the design (flattened into `pcie6_top`)		171,312	5,555,286.020	85.39%
Total	`pcie6_top`	**203,069**	**6,505,672.955**	**100.00%**

Observations.

- Sequential cells are 50,042 of 203,069 instances (24.6%). The paths in the timing report and the instance names in the design-check report show that much of this storage sits in the replay buffer (`u_replay`), the receive FLIT slots (`u_check_mem`) and the depacketizer payload register (`u_depack_pay`). These are wide FLIT-sized buffers and the most direct targets for area reduction (smaller replay window, fewer receive slots, or memory macros).
- The packer is the largest block that stayed separate: 470,968.722, or 7.24% of the area.

- The Tx and Rx CRC pipelines are nearly identical (5,759 and 5,756 cells; about 150,160 area units each, 2.31% of the total each). The CRC is therefore not a dominant cost even at 32 bytes per clock.
- The two PHY blocks together are 2.76% of the area (99,921.841 and 79,160.130), but they dominate timing (Section 8.5).
- The average cell area is about 32.0 area units per instance (6,505,672.955 / 203,069).

8.4 Power

The power report is activity-based (PDB frame `stim#0`) for the whole top module at the 100 MHz clock.

Category	Leakage	Internal	Switching	Total
Register	40.48 μW	338.346 mW	20.702 mW	359.088 mW
Logic	36.49 μW	118.666 mW	64.469 mW	183.171 mW
Memory, latch, clock, pad	0	0	0	0
Total	**76.97 μW**	**457.011 mW**	**85.171 mW**	**542.259 mW**
Share of total	0.01%	84.28%	15.71%	100%

Observations.

- Registers account for 66.22% of the power and combinational logic for 33.78%. The register power is almost entirely internal (clock-pin and cell-internal) power, which follows from having 50,042 flip-flops clocked every cycle.
- Leakage is negligible (about 77 μW). This is expected for a 180 nm-class library; dynamic power is what matters here.
- The clock category is reported as 0 because the clock is ideal at this stage. Clock-tree buffers are not included, so the post-CTS power will be higher than 542.259 mW.
- If, as the instance names suggest, a large share of the flops are in FLIT-sized buffers that are written only once per FLIT, clock gating or enable-based gating of these buffers is the most promising power optimization; this should be confirmed with a per-instance power breakdown.

8.5 Timing

The constraint is a single 10,000 ps clock. The QoR report lists a critical-path slack of 0.0 ps, a TNS of 0.0 and no violating paths. The timing report contains the 40 worst setup paths, all of which are marked MET.

Path group (endpoint)	Paths	Slack (ps)	Cells on path	Startpoint
PHY Tx lane symbol registers (`u_proto_to_phy`)	16	0 to 393	33 to 53	`first`, `s1`
Rx FEC/CRC check: `unc`, `corr_pulse`, `mem`	19	181 to 425	19 to 21	`fec_out_tag`
Replay read pointer (`u_replay_rptr`)	5	408	17	`ack_next`

Critical path (Path 1). The path starts at `u_proto_to_phy/first_reg`, the flag that marks the first beat of a FLIT (it reseeds the scrambler LFSRs and clears the precoder), and ends at `lane3_sym_reg[5]`. It passes through 44 cells. The data path takes 9,654 ps (clock-to-Q alone is 528 ps), the setup time of the capturing flop is 346 ps, and the required time is 9,654 ps, so the slack is 0 ps. The late part of the path is a long alternating chain of XOR and AND-OR cells, which is consistent with the serial dependency inside the mod-4 precoder across the 32 symbols of one lane word together with the 64-step scrambler advance. This is our reading of the cell types and has not been confirmed by a structural analysis of the RTL.

40 worst setup paths. The 40 worst setup paths of the synthesized design (clock period 10,000 ps). Endpoints are shown without the `u_link_layer_` prefix where they belong to the link layer; the PHY endpoints are in `u_proto_to_phy`. "Cells" is the number of combinational cells on the path.

Path	Endpoint (register D pin)	Slack (ps)	Required (ps)	Data path (ps)	Cells
1	u_proto_to_phy/lane3_sym_reg[5]	0	9,654	9,654	44
2	u_proto_to_phy/lane1_sym_reg[3]	0	9,654	9,653	44
3	u_proto_to_phy/lane1_sym_reg[1]	0	9,654	9,653	44
4	u_proto_to_phy/lane3_sym_reg[1]	3	9,528	9,525	40
5	u_proto_to_phy/lane2_sym_reg[5]	3	9,653	9,650	33
6	u_proto_to_phy/lane2_sym_reg[3]	3	9,653	9,650	33
7	u_proto_to_phy/lane2_sym_reg[1]	3	9,653	9,650	33
8	u_proto_to_phy/lane0_sym_reg[5]	5	9,820	9,815	39
9	u_proto_to_phy/lane3_sym_reg[3]	96	9,543	9,448	43
10	u_proto_to_phy/lane0_sym_reg[1]	106	9,543	9,438	53
11	u_proto_to_phy/lane2_sym_reg[7]	152	9,654	9,502	33
12	u_proto_to_phy/lane0_sym_reg[3]	176	9,538	9,362	37
13	u_check_unc_reg[0]	181	9,486	9,305	20
14	u_check_unc_reg[2]	237	9,824	9,587	21
15	u_check_unc_reg[3]	253	9,824	9,571	21
16	u_check_unc_reg[1]	253	9,824	9,571	21
17	u_check_corr_pulse_reg	307	9,784	9,477	21
18	u_proto_to_phy/lane0_sym_reg[7]	315	9,654	9,339	37
19	u_proto_to_phy/lane1_sym_reg[5]	374	9,530	9,156	42
20	u_check_mem_reg[0][176]	378	9,789	9,411	20
21	u_check_mem_reg[0][181]	383	9,789	9,405	20
22	u_proto_to_phy/lane1_sym_reg[7]	384	9,654	9,269	40
23	u_check_mem_reg[0][180]	384	9,789	9,404	20
24	u_check_mem_reg[0][182]	386	9,789	9,402	20
25	u_check_mem_reg[0][183]	387	9,789	9,402	20
26	u_check_mem_reg[0][178]	391	9,789	9,398	20
27	u_check_mem_reg[0][179]	391	9,789	9,398	20
28	u_check_mem_reg[0][177]	391	9,789	9,398	20
29	u_proto_to_phy/lane3_sym_reg[7]	393	9,654	9,260	43
30	u_replay_rptr_reg[6]	408	9,824	9,416	17
31	u_replay_rptr_reg[8]	408	9,824	9,416	17
32	u_replay_rptr_reg[4]	408	9,824	9,416	17

continued on next page

Path	Endpoint (register D pin)	Slack (ps)	Required (ps)	Data path (ps)	Cells
33	u_replay_rptr_reg[5]	408	9,824	9,416	17
34	u_replay_rptr_reg[7]	408	9,824	9,416	17
35	u_check_mem_reg[0][108]	425	9,789	9,363	19
36	u_check_mem_reg[0][105]	425	9,789	9,363	19
37	u_check_mem_reg[0][111]	425	9,789	9,363	19
38	u_check_mem_reg[0][110]	425	9,789	9,363	19
39	u_check_mem_reg[0][104]	425	9,789	9,363	19
40	u_check_mem_reg[0][107]	425	9,789	9,363	19

Reading the timing result.

- All 40 reported paths meet setup at 100 MHz, and the QoR report shows no violating path in the design.
- Eight paths have a slack of 5 ps or less, and all of them end in the PHY lane symbol registers. A slack of 0 ps means the optimizer stopped as soon as the 10 ns target was met; it does not mean the design is exactly at its limit, but it does show that the PHY block has no spare margin at this clock.
- The next group, the Rx FEC correction and CRC-check logic, has 181 to 425 ps of slack, and the replay read pointer has 408 ps. These blocks have 17 to 21 cells on their worst paths, against 33 to 53 for the PHY block.
- The PHY scrambler/precoder is therefore the block that sets the maximum clock rate. A higher clock needs a pipeline cut in the precoder chain (for example, a registered carry between the two 16-symbol halves of the lane word), duplication of the high-fanout `first` flag, or a smaller lane word per clock.

8.6 Design check

The Genus check-design report found no structural problems in the netlist.

Check	Result	Comment
Unresolved references / empty modules	0 / 0	All modules resolved
Assign statements	0	Netlist is fully mapped
Undriven ports / leaf pins	0 / 0	
Multi-driven ports, leaf or hierarchical pins, nets	0	No contention
Unloaded ports, sequential or combinational pins	0	No dead logic left after optimization
Constant pins and ports	0	
Feed-through modules	0	
Undriven hierarchical pins	3	u_packer/s_record[514:512], fanout 0

The three undriven hierarchical pins are the top three bits of the packer's record input. They have no fanout inside the packer, so the netlist is not affected, but the input port can be trimmed or tied

to zero in the RTL to make the check report clean. The report's "logical only instances" entry equals the total instance count (203,069) because no physical (LEF) data has been loaded.

8.7 Limits of these results and next steps

- **Pre-layout numbers.** Timing uses the library's default wireload, so wire delay is only an estimate. Net area is reported as 0. After placement and routing, the zero- and low-slack paths in the PHY block will probably fail at 10 ns until the precoder is pipelined.
- **Ideal clock.** Clock latency and skew are zero and the clock net has a fanout of 50,042 flip-flops. Clock-tree synthesis will add buffers, area and power, and clock skew will affect both setup and hold.
- **Setup only.** The report contains setup checks. Hold checking and fixing belong to the post-CTS stage.
- **Power.** The 542.259 mW figure comes from the activity in one stimulus frame and excludes clock-tree power. It is an estimate for comparing design options, not a sign-off number.
- **Tool flow.** The synthesis was run with the commercial Cadence Genus tool. The problem statement names OpenROAD/OpenLane with open-source PDKs, so the same RTL should also be run through that flow, together with floorplanning, placement, CTS, routing, DRC/LVS and GDSII export.
- **Gate-level simulation.** The mapped netlist `pcie6_mapped_opt.v` has not yet been simulated with the testbench of Section 7. Doing so, with the standard-cell models, would confirm that synthesis preserved the behaviour.

9 Assumptions, Limitations and Future Work

Assumptions where the licensed specification is not reproduced

- The CRC first-consecutive-root FCR is a placeholder (0); the CRC and FEC structure follows public descriptions of the PCIe 6.0 FLIT, not the licensed text.
- The scrambler polynomial and the PAM4 mapping are modelled at symbol level only; there is no analog, equalization or link training.
- Reads and completions: a read is a header-only request record; the testbench models returned data as write-type records in the reverse direction. There is no tag matching between requests and completions and no PCIe ordering rules.

Known limitations

- The transmitter's PHY ready input is tied high in the top level, so there is no PHY backpressure and no link-up sequencing (no LTSSM).
- PCIe credit-based flow control is not modelled; the replay window is the only flow control.
- tqos is read from the master pins when arbitration happens rather than stored with the beat. There is no starvation guard or lock timeout in the interconnect, so a master that never finishes a packet holds the lock.
- Records are not split across FLITs, so a FLIT may carry padding; the 24-bit reserved field of the DLP is unused.
- The synthesized design meets timing at 100 MHz, a tenth of the full PCIe 6.0 per-lane rate. The PHY scrambler/precoder has the least timing margin (0 ps worst slack, Section 8.5), so a higher clock needs pipelining there; the parallel CRC and FEC logic has 181 ps or more of slack at this clock.

- The synthesis results are pre-layout (default wireload, ideal clock, setup checks only) and were produced with a commercial tool rather than the open-source flow named in the problem statement.

Future work

OpenLane/OpenROAD synthesis, place-and-route and GDSII; post-layout timing with CTS and hold fixing; pipelining of the PHY precoder and scrambler to raise the clock rate; reduction of the FLIT-sized buffers (replay window, receive slots, depacketizer register) and clock gating to cut area and power; gate-level simulation of the netlist; FPGA bring-up (bonus deliverable); SVA and Python golden model; credit flow control and link training; per-beat QoS storage with aging; tag-based completions.

10 Conclusion

We built a full-duplex AXI4-Stream to PCIe 6.0-style bridge that follows the given three-layer block diagram and adds what a real link needs: packet-atomic QoS arbitration, a lossless metadata scheme with a TLP-style header and a DLP, CRC and FEC protection with a receive-side decoder, sequence-numbered ACK/NAK and replay, and a lane mapping with byte striping, scrambling, Gray coding and precoding. A two-instance testbench with a scoreboard and error injection covers the network-packet and ML-inference workloads. Logic synthesis of the whole design with Cadence Genus at 100 MHz gives 203,069 cells with 6,505,672.955 area units and an estimated 542.3 mW, with all reported setup paths met and a clean netlist check; the PHY scrambler/precoder is the timing-critical block and the FLIT-sized buffers dominate storage and power. Remaining work is the recorded simulation results, place-and-route and GDSII in the open-source flow, post-layout timing, and the FPGA bonus.

References

1. PCI-SIG, *PCI Express Base Specification, Revision 6.0* (licensed document; FLIT mode, FEC, CRC and replay).
2. Arm, *AMBA AXI-Stream Protocol Specification* (IHI 0051).
3. Arm, *AMBA AXI and ACE Protocol Specification* (IHI 0022), for the AxQOS priority concept.
4. I. S. Reed and G. Solomon, Polynomial Codes over Certain Finite Fields, *Journal of the Society for Industrial and Applied Mathematics*, 1960.
5. Cadence Design Systems, *Genus Synthesis Solution*, release 20.11 (synthesis, area, power, timing and check-design reports used in Section 8).
6. Project source files: pcie6_top.v, axi4s_slave_if.v, axi_interconnect.v, pcie6_link_layer.v, pcie6_flit_protect.v, pcie6_rx.v, pcie6_phy.v and tb_pcie6_link.v.
7. Synthesis outputs: pcie6_mapped_opt.v (netlist), area.rpt, power.rpt, qor.rpt, timing.rpt and design_check.rpt.